Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Electrostatic Discharge Control (ESD) Masterclass

Course Code: C337

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	Initial C337 release of the ESD control courseware, delivered as a one-day 9:30am – 5:30pm course (7.5 instructional hours).	Tertiary Infotech Academy Pte Ltd

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Course Overview

One-day course, 9:30am – 5:30pm (7.5 instructional hours), in four modules: ESD fundamentals and failure mechanisms; ESD control principles, equipment and practices; ESD standards, testing and auditing; and training, awareness and incident improvement. Eight applied activities run through the day.

Learning Outcomes

- LO1: Apply control measures to manage ESD generation based on ESD principles.
- LO2: Evaluate and recommend actions to improve ESD control effectiveness in accordance with ESD standards.

Course Schedule · 9:30am – 5:30pm

Time	Session / learning activity	Slide reference
09:30–09:45	Welcome, introductions, ground rules and course overview	1–8
09:45–10:20	Module 1: charge generation, discharge paths and models	9–23
10:20–11:10	Failure modes, exposure points and business impact	24–35
11:10–11:35	Activities 1–2: hazard hunt and failure triage	Act 1: 36–38, Act 2: 39–41, recap 42
11:35–12:25	Module 2: EPA, grounding, personnel and worksurface controls	43–53
12:25–12:55	Lunch	54
12:55–13:35	Ionisation, shielding, packaging and material movement	55–64
13:35–14:10	Activities 3–4: EPA design and automated-cell controls	Act 3: 65–67, Act 4: 68–70, recap 71
14:10–14:50	Module 3: standards and programme architecture	72–77
14:50–15:35	Measurement discipline, instruments and verification	78–88
15:35–16:10	Activities 5–6: verification round and standards audit	Act 5: 89–91, Act 6: 92–94, recap 95
16:10–16:45	Module 4: competence, incident response and CAPA	96–108
16:45–17:05	Activity 7: incident investigation	Act 7: 109–111
17:05–17:20	Activity 8: management capstone and course summary	Act 8: 112–114, 115–116
17:20–17:30	Course recap, Q&A and course feedback	117–118

Activity Alignment

Activity	Module	Output	Slides
1. Static Hazard Hunt at NovaSense Incoming Inspection	Module 1	Annotated hazard map; Risk-prioritisation worksheet; Three-	36–38

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		control proposal; Peer-review note	
2. Medical Microchip Failure Triage: ESD, EOS or Process Defect?	Module 1	Chronology; Hypothesis matrix; Containment plan; Decision-gate briefing	39–41
3. Design and Commission an EPA for Orion Optics	Module 2	EPA layout; Grounding/bonding diagram; Insulator-control register; Commissioning plan and readiness certificate	65–67
4. Select Controls for the Aster Robotics Handling Cell	Module 2	Exposure-path map; Control-selection matrix; Monitoring/stop-condition plan; Maintenance plan	68–70
5. Execute a Compliance Verification Round	Module 3	Completed measurement log; Failure/disposition records; Metric calculation; Signed evidence package	89–91
6. Standards-Based EPA Audit at Helix Assembly	Module 3	Audit plan; Completed checklist; Evidence register; Graded findings and follow-up plan	92–94
7. Investigate the PulseMed ESD Incident	Module 4	Containment record; Timeline and causal analysis; CAPA plan; Effectiveness-verification plan	109–111
8. Management Capstone: Build the Apex ESD Control Plan	Modules 1–4	ESD control plan; Qualification and verification matrices; Competency matrix; Metrics dashboard and 90-day roadmap	112–114

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